

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	N. SRIKANTH	Reg. No.:	192411139
Section / Batch:	CSE	Date:	16/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Multi -Source Log Processing System

A distributed log processing platform divides input into two equal parts and performs logarithmic work during the merging phase due to search and synchronization operations. Apply the substitution method to solve the recurrence $T(n)=2T(n/2)+\log n$. Expand the recurrence in detail and derive the final complexity. Analyze performance for large log volumes. Interpret the result using asymptotic notation.

Q2. Hierarchical Tree Query Engine

A tree-processing engine reduces the problem size by half at each recursive level and performs logarithmic work for query evaluation. Apply the substitution method to solve the recurrence $T(n)=T(n/2)+\log n$. Expand the relation step-by-step and explain the recursion depth. Derive the final asymptotic complexity and analyze its scalability. Interpret the result using asymptotic notation.

Code of Conduct

I N. Srikanth (192411139) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

ANSWER:

Q1. Given the recurrence:

$$T(n) = 2T\left(\frac{n}{2}\right) + \log n$$

We will solve it using the substitution (expansion) method.

Step 1: Expand the recurrence

Given,

$$T(n) = 2T\left(\frac{n}{2}\right) + \log n$$

Expand $T(n/2)$:

$$T\left(\frac{n}{2}\right) = 2T\left(\frac{n}{4}\right) + \log\left(\frac{n}{2}\right)$$

Substitute back:

$$\begin{aligned} T(n) &= 2 \left[2T\left(\frac{n}{4}\right) + \log\left(\frac{n}{2}\right) \right] + \log n \\ &= 4T\left(\frac{n}{4}\right) + 2\log\left(\frac{n}{2}\right) + \log n \end{aligned}$$

Expand again:

$$T\left(\frac{n}{4}\right) = 2T\left(\frac{n}{8}\right) + \log\left(\frac{n}{4}\right)$$

Substitute:

$$T(n) = 8T\left(\frac{n}{8}\right) + 4\log\left(\frac{n}{4}\right) + 2\log\left(\frac{n}{2}\right) + \log n$$

After k expansions,

$$T(n) = 2^k T\left(\frac{n}{2^k}\right) + \sum_{i=0}^{k-1} 2^i \log\left(\frac{n}{2^i}\right)$$

Step 2: Reach the base case

The recursion stops when

$$\frac{n}{2^k} = 1$$

Therefore,

$$2^k = n$$

and

$$k = \log n.$$

Substituting,

$$T(n) = nT(1) + \sum_{i=0}^{\log n - 1} 2^i \log\left(\frac{n}{2^i}\right).$$

Since

$$\log\left(\frac{n}{2^i}\right) = \log n - i,$$

we get

$$T(n) = nT(1) + \sum_{i=0}^{\log n - 1} 2^i (\log n - i).$$

Step 3: Evaluate the summation
Separate the sum:

$$T(n) = nT(1) + \log n \sum_{i=0}^{\log n - 1} 2^i - \sum_{i=0}^{\log n - 1} i 2^i.$$

Now,

$$\sum_{i=0}^{m-1} 2^i = 2^m - 1 = n - 1 = \Theta(n),$$

and

$$\sum_{i=0}^{m-1} i 2^i = \Theta(n \log n).$$

Thus,

$$T(n) = \Theta(n) + \Theta(n \log n) - \Theta(n \log n) = \Theta(n).$$

The $n \log n$ terms cancel, leaving only a linear term.

Verification by substitution

Assume

$$T(n) \leq cn - d \log n.$$

Substitute into the recurrence:

$$\begin{aligned} T(n) &= 2 \left[c \frac{n}{2} - d \log\left(\frac{n}{2}\right) \right] + \log n \\ &= cn - 2d(\log n - 1) + \log n \\ &= cn - (2d - 1) \log n + 2d. \end{aligned}$$

Choosing suitable constants c and d satisfies the induction hypothesis, confirming

$$T(n) = \Theta(n).$$

Final Complexity

$$\boxed{T(n) = \Theta(n)}$$

Q2. Given the recurrence:

$$T(n) = T\left(\frac{n}{2}\right) + \log n$$

we solve it using the substitution (expansion) method.

$$T(n) = T\left(\frac{n}{2}\right) + \log n$$

Step 1: Expand the recurrence

Start with

$$T(n) = T\left(\frac{n}{2}\right) + \log n$$

Expand $T(n/2)$:

$$T\left(\frac{n}{2}\right) = T\left(\frac{n}{4}\right) + \log\left(\frac{n}{2}\right)$$

Substitute back:

$$T(n) = T\left(\frac{n}{4}\right) + \log\left(\frac{n}{2}\right) + \log n$$

Expand once more:

$$T\left(\frac{n}{4}\right) = T\left(\frac{n}{8}\right) + \log\left(\frac{n}{4}\right)$$

Substitute again:

$$T(n) = T\left(\frac{n}{8}\right) + \log\left(\frac{n}{4}\right) + \log\left(\frac{n}{2}\right) + \log n.$$

After k expansions,

$$T(n) = T\left(\frac{n}{2^k}\right) + \sum_{i=0}^{k-1} \log\left(\frac{n}{2^i}\right).$$

Step 2: Determine the recursion depth

The recursion ends when

$$\frac{n}{2^k} = 1.$$

Hence,

$$2^k = n$$

which gives

$$k = \log n.$$

Therefore, the recursion depth is

$$\boxed{\log n}.$$

Substituting $k = \log n$,

$$T(n) = T(1) + \sum_{i=0}^{\log n - 1} \log\left(\frac{n}{2^i}\right).$$

Since

$$\log\left(\frac{n}{2^i}\right) = \log n - i,$$

we obtain

$$T(n) = T(1) + \sum_{i=0}^{\log n - 1} (\log n - i).$$

Step 3: Evaluate the summation

Let $m = \log n$.

Then,

$$\begin{aligned} T(n) &= T(1) + \sum_{i=0}^{m-1} (m - i) \\ &= T(1) + (m + (m - 1) + \cdots + 1). \end{aligned}$$

The sum of the first m positive integers is

$$\frac{m(m + 1)}{2}.$$

Hence,

$$T(n) = T(1) + \frac{\log n(\log n + 1)}{2}.$$

Ignoring constants and lower-order terms,

$$T(n) = \Theta((\log n)^2).$$

Verification by substitution

Assume

$$T(n) \leq c(\log n)^2.$$

Substitute into the recurrence:

$$\begin{aligned}T(n) &\leq c(\log(n/2))^2 + \log n \\&= c(\log n - 1)^2 + \log n \\&= c(\log n)^2 - 2c\log n + c + \log n.\end{aligned}$$

For sufficiently large n , the lower-order terms are dominated by $c(\log n)^2$, confirming the assumption.

Thus,

$$\boxed{T(n) = \Theta((\log n)^2)}.$$

Recursion Depth

- Problem size is halved at every recursive call.
- Number of recursive levels:

$$\boxed{\log n}$$

- At level i , the work performed is

$$\log\left(\frac{n}{2^i}\right) = \log n - i.$$

The total work is the sum of these logarithmic costs over all $\log n$ levels, giving a quadratic logarithmic growth.

RUBRICS FOR CALCULATION

Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, wellstructured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____